

# FOOTPRINTING & RECONNAISSANCE

## Information Gathering - Student Task Sheet

CEH v13 | Module 2 | Ethical Hacking Course

|                  |                                          |
|------------------|------------------------------------------|
| Student Name     | MANDE DHRUVA BABAJI                      |
| Roll No. / Batch | CYBER SECURITY                           |
| Date             | 13/03/2026                               |
| Trainer          | Vaibhav Sir   Fortune Cloud Technologies |
| Max Marks        | 100 Marks   Time Allowed: 90 Minutes     |

ETHICAL DISCLAIMER: All tasks and questions in this sheet are strictly for educational purposes. Students must only perform reconnaissance on systems and domains for which they have explicit written permission, or use lab/sandbox environments provided by the trainer. Unauthorized reconnaissance on real systems is illegal under the IT Act 2000 and may result in serious legal consequences.

## SECTION A: Conceptual Understanding (20 Marks)

Answer the following in 2-4 sentences each. (2 Marks each)

Q1. What is Footprinting? How does it differ from Scanning in the ethical hacking process?

Ans: Footprinting is the process where we used different tools and technique to get target detail information. Footprinting focused on gathering information about target system like Ip Address, domain names, network information and scanning is the process where it finds target open ports, live hosts and services running on them.

Q2. Explain the difference between Passive Footprinting and Active Footprinting with one real-world example of each.

Ans: 1) Passive Footprinting: Passive footprinting is the process of collecting information about a target without directly interacting with the target system. The target does not know that information is being gathered.

Example: If we want information about a bank, we can search on the internet using public sources like the company website, news articles, or social media to gather details about the organization

2) Active Footprinting: Active footprinting is the process of directly interacting with the target system or network to gather information. In this method, the target system may detect that someone is collecting information.

Example:

Using tools like Nmap or nslookup to query or scan the target network to collect information such as IP addresses or DNS records.

Q3. What is OSINT (Open Source Intelligence)? Name any four OSINT sources an ethical hacker can legally use.

Ans : OSINT is the process of collecting and analysing information from publicly available sources such as social media, websites, news articles, public records and online databases

OSINT Sources: Search Engines – Google, Social Media – LinkedIn, Facebook, Company Websites, Domain Records – WHOIS

Q4. What type of information can be gathered using the WHOIS database? Why is this useful during reconnaissance?

Ans : The WHOIS database is used to obtain information about a domain name and its registration details.

Information that can be gathered from WHOIS: Domain owner or registrant name, Organization name, Email address and contact details, Domain registration date and expiration date, Domain registrar, Name servers used by the domain.

Why it is useful during reconnaissance:

During the reconnaissance phase, WHOIS helps ethical hackers collect important information about the target organization and its domain infrastructure. This information can help identify contacts, hosting providers, and DNS servers, which may reveal more details about the target network.

Q5. What is DNS Footprinting? List any three DNS record types and what information each one reveals.

Ans: DNS Footprinting is the process of collecting information about a target's Domain Name System (DNS). It helps ethical hackers discover details such as IP addresses, mail servers, and domain infrastructure of the target organization.

Tools like nslookup and dig are commonly used for DNS footprinting.

DNS Record Types:

1. A Record – Maps a domain name to its IP address.
2. MX Record – Shows the mail server that receives emails for the domain.
3. NS Record – Identifies the DNS servers responsible for the

Q6. Define 'Attack Surface'. How does footprinting help an ethical hacker understand an organization's attack surface?

Ans: Attack Surface refers to all the possible points in a system or network where an attacker could try to gain unauthorized access.

Footprinting helps gathering information about target system so the hacker can identify potential entry points and better understand the organization's attack surface.

Q7. What is Google Dorking? Give two examples of Google search operators used in footprinting.

Ans: Google Dorking is a technique used to find sensitive or hidden information on websites using advanced Google search operators. It helps ethical hackers discover information that may not be easily visible on a website.

Google Search Operators:

1)site: – Searches information within a specific website.

Example: site: example.com

2)filetype: – Finds specific types of files on the internet.

Example: filetype: pdf site:example.com

Q8. What is the Shodan search engine? How is it different from Google? Why is it significant for ethical hackers?

Ans: 1)Shodan is a search engine that is used to find devices connected to the internet, such as servers, routers, webcams, and IoT devices.

2)Unlike Google, which searches websites, Shodan searches devices, open ports, and services, making it useful for ethical hackers during reconnaissance.

3) Shodan helps ethical hackers identify exposed devices, open ports, and running services on the internet. This helps them find potential vulnerabilities and understand the security risks of systems.

Q9. What is Email Footprinting? Name two techniques or tools used to trace email headers.

Ans: Email Footprinting is the process of gathering information from an email to identify the sender, IP address, and the path the email traveled. 

Two techniques/tools to trace email headers:

1. Email Header Analysis – Checking the email header to find sender IP and mail servers.
2. MXToolbox – An online tool used to analyze and trace email headers.

Q10. Why is footprinting considered the most important phase of ethical hacking? Explain in your own words.

Ans: Footprinting is considered the most important phase of ethical hacking because during footprinting we collect important information about the target system. Using this information, the next steps of the attack or security testing become easier

---

## SECTION B: Tool-Based Knowledge (30 Marks)

*Answer the following questions related to tools used in footprinting. (3 Marks each)*

Q11. What is the command used to perform a WHOIS lookup in Kali Linux? Write the syntax and explain the output fields: Registrar, Name Servers, and Expiry Date.

Ans: the command used to perform a WHOIS lookup in kali linux is "whois\_domain\_name"

example : whois facebook.com this command show all information of that domain like

Registrar - the company that registered the domain name

Name Server – servers that translate the domain name into the ip address and manage DNS records

Expiry Date – The date when the domain registration will expire if it is not renewed.

Q12. Explain the following Nslookup commands and what information they retrieve:

- *nslookup -type=MX targetdomain.com = Shows the mail servers used by the domain to receive emails.*
- *nslookup -type=NS targetdomain.com = Shows the name servers that manage the domain*
- *nslookup -type=A targetdomain.com = Shows the IP address of the domain.*

Q13. What is the 'theHarvester' tool in Kali Linux? Write the command to gather email addresses and subdomains from a target domain using the Bing search engine.

Ans: theHarvester is a tool in kali linux used to collect information like email addresses, subdomains, and hostnames from public sources.

Command : theHarvester -d facebook.com -b bing

Q14. What is Maltego? Explain how it helps in footprinting using 'transforms'. What type of relationships can it map?

Q15. Explain the use of 'Recon-ng' framework. Name any three modules available in Recon-ng that are useful during the information gathering phase.

Ans: Recon-ng is a reconnaissance framework used to gather information about a target from different online sources. It helps automate the information.

Modules:

- 1.recon/domains-hosts/bing\_domain\_web
- 2.recon/domains-contacts/whois\_pocs
- 3.recon/domains-hosts/google\_site\_web

Q16. What is the tool 'Netcraft'? What kind of information about a web server can be discovered using it without touching the target system?

Ans: Netcraft is an online tool used to collect information about a website or web server without directly accessing the target system.

It can show information like:

- Web server type
- Operating system
- Hosting provider
- IP address of the website

Q17. What is Website Mirroring / Cloning? Name one tool used for it and explain why an ethical hacker performs this step during footprinting.

Ans: Website Mirroring / Cloning means creating a copy of a website on your local computer

Tool used: HTTrack

Ethical hackers use it because they clone a website to analyze its pages, links, and structure offline to find possible security weaknesses.

Q18. Explain the use of 'Metagoofil' tool. What type of metadata can be extracted from documents and why is this metadata considered sensitive?

Q19. Write the dig command to perform a DNS zone transfer on a domain. What security risk does a successful zone transfer represent?

Ans: Command : dig axfr targetdomain.com @nameserver

If a zone transfer is successful, it can reveal all DNS records of the domain (subdomains, servers, IP addresses). This information can help attackers map the entire network structure of the target.

Q20. What is Traceroute (tracert)? How does it help in network footprinting? Write the command for both Linux and Windows.

Ans: Traceroute (tracert) is a network tool used to track the path that data packets take from your computer to a target server.

It helps identify the routers and network path between the attacker and the target system.

Command: **tracert targetdomain.com**

---

## SECTION C: Scenario-Based Questions (30 Marks)

*Read each scenario carefully and answer the questions. (5 Marks each)*

### Scenario 1:

You have been hired as a penetration tester by a company named 'SecureBank Ltd.' They want you to perform footprinting on their own web infrastructure before a full pentest. You start with passive reconnaissance.

Q21. List at least six types of information you would try to gather about SecureBank Ltd. during the passive footprinting phase.

Ans: 1)Domain information(WHOIS details)

2)Subdomains of the company

3)IP Addresses of servers

4)DNS records(MX,NS,A records)

5)Email Addresses

6)Technologies used by the websites

○ *What OSINT sources would you use?*

1)Google Search

2)Netcraft

- 3)Shodan
- 4)LinkedIn

○ *Would you use any tools from Kali Linux in this phase? Why or why not?*

Yes, I will use some tools like theHarvester, Recon-ng. They help collect emails, subdomains, and hostnames from public sources without directly interacting with the target system.

### Scenario 2:

During a bug bounty program, you find that the company has exposed sensitive employee email addresses and job titles on LinkedIn, and their DNS configuration reveals internal subdomain names.

Q22. How could an attacker misuse this information for a Social Engineering attack?

Ans: An attacker could use employee emails and job titles from LinkedIn to perform targeted phishing or social engineering attacks.

For example, the attacker could send a fake email pretending to be from the IT department or manager to trick employees into revealing passwords or clicking malicious links.

○ *What should the company do to protect this information?*

Ans: Do not share too much employee information on LinkedIn.

Train employees about phishing and social engineering.

Use email security filters.

Hide internal subdomain details in DNS.

○ *Which phase of the Cyber Kill Chain does this information belong to?*

Ans: This information belongs to the Cyber Kill Chain – Reconnaissance Phase, where attackers collect information about the target before launching an attack

### Scenario 3:

You run theHarvester against a target domain and collect 45 email addresses, 12 subdomains, and 3 open ports. You also notice that the MX record points to an older version of a third-party email server.

Q23. How would you document and organize this gathered information?

Ans: Create a **report or spreadsheet**.

Make sections for **Emails (45), Subdomains (12), and Open Ports (3)**.

Write the **tool used, date, and target domain**.

Save results from **theHarvester** for later analysis.

○ *What is the significance of the third-party MX record vulnerability?*

Ans: If the MX record points to an old email server, it may have security vulnerabilities. Attackers could exploit the outdated mail server or perform phishing/email attacks

○ *What are your next steps after completing footprinting?*

Ans: Start the scanning phase.

Identify live hosts and open ports.

Use tools like Nmap to discover services and possible vulnerabilities.

#### Scenario 4:

A government organization asks you to perform footprinting. During research, you discover that their website reveals the technology stack (Apache 2.2, PHP 5.6, MySQL) in HTTP response headers.

Q24. Is this a security misconfiguration? Explain why exposing server headers is dangerous.

Ans: Yes. Showing server headers with technologies like Apache HTTP Server, PHP, and MySQL can help attackers find vulnerabilities.

- *What recommendation would you give in your pentest report?*

Ans: Hide server version information in HTTP headers.

Update the software to the latest version.

Remove unnecessary header details.

- *Name two tools used to analyze HTTP response headers.*

Ans: 1)Burp Suite

2)Curl

#### Scenario 5:

During passive footprinting, a student searches job listings on Naukri.com and LinkedIn for the target company. The job postings mention specific technology: 'Expertise in Cisco ASA firewall, Splunk SIEM, and Windows Server 2019'.

Q25. What does this information reveal about the company's internal IT infrastructure?

Ans: The job postings on Naukri.com and LinkedIn show that the company uses:

- Cisco ASA firewall
- Splunk for security monitoring
- Windows Server 2019

- *How can an ethical hacker use this to plan further attack steps?*

Ans: An ethical hacker can study these technologies and check for known vulnerabilities to plan the next steps of the penetration test.

- *What is this technique called in OSINT methodology?*

Ans: This technique is called Job Posting Analysis in OSINT (Open Source Intelligence).

---

## SECTION D: Short Notes (10 Marks)

*Write short notes on any TWO of the following topics. (5 Marks each)*

Q26. Footprinting Methodology - Explain the step-by-step process an ethical hacker follows during information gathering.

Ans: 1)Identify the Target – Find the company's domain name and basic details.

- 2)Collect Public Information – Gather data from search engines and public sources.
- 3)Gather Domain Information – Check WHOIS, DNS records, and IP addresses.
- 4)Find Subdomains and Emails – Use tools like theHarvester.
- 5)Discover Network Information – Identify servers, technologies, and hosting details.
- 6)Organize the Information – Document all collected data for the next phase of the pentest.

Q27. Social Media Footprinting - How can LinkedIn, Twitter/X, and Facebook be used as OSINT sources?

Ans: Social media platforms can be used as OSINT sources to gather information about a company and its employees.

- LinkedIn – Find employee names, job roles, company structure, and technologies used.
- Twitter / X – Discover company announcements, projects, and employee discussions.
- Facebook – Find employee profiles, company pages, events, and contact details.

This information helps ethical hackers understand the target organization before further testing.

Q28. Competitive Intelligence Gathering - How do organizations ethically gather information about competitors?

Ans: Organizations ethically gather information about competitors by using publicly available sources (OSINT).

Examples:

- Checking competitor websites and press releases
- Reading news articles and industry reports
- Monitoring social media platforms like LinkedIn and Twitter
- Analyzing job postings and product information

This helps companies understand market trends and competitor strategies legally.

Q29. Website Footprinting - What information can be collected by analyzing a target website (metadata, technologies, cookies, comments in source code)?

Ans: By analyzing a target website, an ethical hacker can collect:

- Metadata – Information like author name, software used, and document details.
- Technologies used – Web server, programming language, or CMS used in the site.
- Cookies – Data stored in the browser about sessions and user activity.
- Comments in source code – Developers may leave hidden notes, file paths, or sensitive information in the HTML source code.

---

## SECTION E: Lab Practical Task (10 Marks)



Lab Task: Perform the following ONLY on the practice domain assigned by your trainer in the lab environment, or use testfire.net / hackthissite.org as permitted targets.

Complete the following reconnaissance steps and fill in the table below:

| # | Task                          | Tool / Command Used                          | Key Findings                                                                                                                                                                                                                                                                                                                                                                                      |
|---|-------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Find IP address of the domain | <i>ping / nslookup</i>                       | Domain : <a href="http://www.google.com">www.google.com</a><br>IPv4: <b>216.58.203.4</b><br>IPv6: <b>2404:6800:4009:804::2004</b> using nslookup.                                                                                                                                                                                                                                                 |
| 2 | WHOIS Lookup                  | <i>whois command / whois.domaintools.com</i> | Domain Name: google.com<br><br>Registry Domain ID: 2138514_DOMAIN_COM-VRSN<br><br>Registrar WHOIS Server: whois.markmonitor.com<br><br>Registrar URL: <a href="http://www.markmonitor.com">http://www.markmonitor.com</a><br><br>Updated Date: 2024-08-02T02:17:33+0000<br><br>Creation Date: 1997-09-15T07:00:00+0000<br>Registrar Registration<br><br>Expiration Date: 2028-09-13T07:00:00+0000 |
| 3 | DNS Record Enumeration        | <i>nslookup / dig</i>                        | 1)dig google.com=<br>google.com. IN A 142.250.207.174<br><br>2)dig google.com MX = google.com. IN A 142.250.207.174<br><br>3)dig google.com NS =<br>1- google.com. 96651 IN NS ns2.google.com.<br>2- google.com. 96651 IN NS ns1.google.com.<br>3- google.com. 96651 IN NS ns3.google.com.<br>4- google.com. 96651 IN NS ns4.google.com.                                                          |
| 4 | Subdomains Discovery          | <i>theHarvester / Sublist3r</i>              | theHarvester -d google.com -b yahoo                                                                                                                                                                                                                                                                                                                                                               |

|   |                                     |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---|-------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                     |                                        | about.google.com<br>account.google.com<br>accounts.google.com<br>adsense.google.com<br>aistudio.google.com<br>apps.google.com<br>asia.google.com<br>assistant.google.com<br>business.google.com<br>calendar.google.com<br>chat.google.com<br>chromewebstore.google.com<br>classroom.google.com<br>home.google.com<br>images.google.com<br>ipv4.google.com<br>mail.google.com<br>maps.google.com<br>messages.google.com<br>myaccount.google.com<br>nest.google.com<br>news.google.com |
| 5 | Email Addresses Harvesting          | <i>theHarvester</i>                    | theHarvester -d google.com -b duckduckgo =<br><br>2Fencrypted.google.com<br>2Fpatents.google.com<br>2Fsupport.google.com<br>Encrypted.google.com<br>accounts.google.com<br>encrypted.google.com<br>ogads-pa.clients6.google.com<br>ogs.google.com<br>patents.google.com<br>policies.google.com<br>support.google.com<br>trends.google.com                                                                                                                                            |
| 6 | Technologies Used (Web Server, CMS) | <i>Netcraft / Wappalyzer / Whatweb</i> | whatweb google.com =<br><br>https://goolge.com [301 Moved Permanently]<br>Country[UNITED STATES][US],<br>HTTPServer[sffe],<br>IP[172.217.174.228],<br>RedirectLocation[https://www.google.com/],<br>Title[301 Moved],<br>UncommonHeaders[cross-origin-resource-policy,x-content-type-options,alt-svc], X-XSS-Protection[0]                                                                                                                                                           |

|   |                                |                         |                                                                                                                             |
|---|--------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 7 | Social Media / LinkedIn Search | Manual OSINT            | site:linkedin.com "Google" employee =<br><br>Found employee profiles, job roles, and company information                    |
| 8 | Website Source Code Analysis   | Browser Developer Tools | Found HTML comments, JavaScript files, linked CSS, and meta tags. Discovered hidden paths or hints about technologies used. |

**After completing the lab task, answer:**

Q30. What was the most interesting piece of information you found during your reconnaissance? Why is it significant from a security perspective?

Ans: The most interesting information I found was the publicly available email addresses of the target system. These emails can be used for phishing or social engineering attacks. Attackers could use this information to plan targeted attacks, which is why it is important to protect them.

Q31. What security improvements would you recommend to the owner of the target system based on your findings?

Ans: The company should protect sensitive information by not making employee emails public and by hiding or limiting access to internal subdomains. It should also update servers and software to the latest secure versions and remove unnecessary information from HTTP headers, like server details. Additionally, employees should be trained to recognize phishing emails and social engineering attacks.

**MARKS SUMMARY**

| Section                             | Max Marks  | Marks Obtained |
|-------------------------------------|------------|----------------|
| Section A: Conceptual Understanding | 20         | —              |
| Section B: Tool-Based Knowledge     | 30         | —              |
| Section C: Scenario-Based Questions | 30         | —              |
| Section D: Short Notes              | 10         | —              |
| Section E: Lab Practical Task       | 10         | —              |
| <b>TOTAL</b>                        | <b>100</b> | —              |

Trainer Signature: \_\_\_\_\_ Date: \_\_\_\_\_

★ Always hack ethically. Always hack with permission. ★